



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8397**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	305	Seat No:	2023701010
Course Section:	1				



Name of the candidate. Ajit Yadav

Candidate's UID number:

2	0	2	3	7	0	1	0	1	0
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Examination class room number:

3	0	5
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Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE - DS

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 08/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q1.	
	(a)	$f(w) = \frac{1}{2} \ Xw - y\ ^2$ Gradient- update rule are use to find the best weight for perceptron. The error is calculated by actual output and predicted one then it is updated. We find the new weight- then we calculate the error i.e. $(y - \bar{y})$ and update the new weight. $w_{new} = w_{old} + \alpha \cdot (y - \bar{y}) \cdot X$ where α - is learning rate If X is old input value we continue till the error is minimized.

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(Begin answer for each question on a new page)

Q1.

b. Bias - Variance trade off

High bias mean the error is more between the actual and predicted one, Variance is used to measure how well the model perform for new dataset of model.

Overfitting occurs when model capacity is too high it affects generalization.

Variance is measure how the model perform when it is test on the new data, Calculate the difference between actual and predicted on with help of Bias-variance tradeoff.

When the Bias is high and variance is low it means it perform well during test but during training there was huge difference between actual and predicted data.

When Bias is low and variance is high. This means it perform well during training but the test time it was not well performed as there was huge difference between actual and predicted.

High bias can cause overfitting as well low bias can cause underfitting. So, the bias and variance should be medium.

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Q.

2.a.

2D convolution is a depthwise convolution for input size $H \times W \times C$ with K filters of size $F \times F$.

In 2D Convolution the convolution is done with 2D grid of $F \times F$ where multiple filter are convoluted to find the edge, max, shapes, etc while in depth convolution the filter is placed on the grid and then operation occurred like to find the max or min for particular cell and the filter ~~se~~ cell is moved by one and the same process occur till $n-1$ rows and then the filter columns move toward down and the same process occur till $n-1$ row.

The filter parameter is used to upon the 2D grid and then for multiple filter and then the parameter like weight and bias is updated by find error between actual and predicted one the weight is updated.

Each pixel values is taken as input and multiple filter perform as hidden ~~latter~~ layer and then it provided the output.

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Q.

2.b. The linear under complete auto encoder trained with MSE (Mean Squared Error) learns a subspace equivalent to PCA.

In PCA the dimensionality reduction technique. we find the matrix by compute mean then we find the eigenvalue and eigenvector.

The dimensionality reduction is done with help of eigenvector found.

The linear under complete auto encoder trained with MSE learns subspace equivalent to PCA as it find the linearity between two field or columns and it give the linear output between them that is nothing but the projection between two attribute with new dimensionality that is combination of both.

The auto encoder find the vector space related to it with help of word to vector format.

The over completed auto encoder without regularization tends to learn the identity function as the encoder is completed and it has encoded all portion in equivalent space i.e word to vector form.



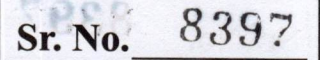
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Q.

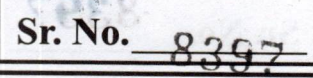
3a. RNN ~~(Recurrent)~~

Recursive gradient expression $\frac{\partial L}{\partial h_{t-k}}$

RNN is used for translation where it does not store the previous data. It merge the vector with new word and encode it and keep doing this process then the decoding is also done by same manner but previous word it does not have so the meaning difficult to capture where LSTN (Long Short Term Memory) it ~~sto~~ save the and capture the previous word.

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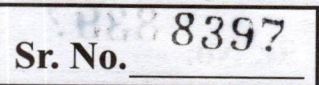
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